Element D:
Design Concept Generation, Analysis, And Selection

BEST SOLUTIONS

For this problem, our best solution is mirrors, more specifically, convex ones, since they increase the field of view for each mirror. We decided this using a decision matrix, which eliminated all of our other ideas and left mirrors as the least expensive, the most variable, and the most easily-assembled design. We chose convex mirrors because the area that we want to specifically target with this is the corners of the library because it is one of the biggest blindspots for the librarian to monitor. By making this a dome shape helps us have a broad vision of the corner to make sure that we have a large surface area to have a 180 degree view of the area.

BRAINSTORMING IDEAS/SOLUTIONS

The ideas — other than mirrors — include translating the librarian's table to the center of the library, allowing for a 180 degree view from her desk, the initial problem that we had to solve. Another idea includes tearing down the bookshelves that were causing blind spots in the first place, but looking back, it was a really bulky idea that would've required an immense amount of time and effort. One of the last ideas that we came up with as a class was setting up security cameras around the library that showed video feeds of the blind spots of the library on a monitor that the librarian would be able to see. This was an idea that would've worked very well, but it would've required a lot of work with wires, electricity, which seemed like too much of a hassle compared to mirrors. The last idea that we presented as a class was hanging the librarian's desk from the ceiling. It would've presented the librarian with a top-down view of the library, above any top of any bookshelf in the library. From there, she would be able to see the entire library at once by just looking down. This was a silly idea in retrospect, but during the brainstorming session, all ideas are allowed since they may be the innovation of tomorrow.

DECISION

The class decided to place convex curved mirrors in each of the upper corners of Eleanor Roosevelt High School's library. To decide, we used a decision matrix which narrowed down the ideas and showed that mirrors would be the most efficient because of its ability to reflect 180 degrees making it effective for corners and most cost effective. This is not restrained by quality because our design is only needed to see whether there are any students in the corner and not whose in the corner. To ensure that others can replicate our results we wrote out our precise measurements and dimensions and gave a scaled down sketch of all the parts. For more vague portions they can look at our material lists and documents to understand what we did.

Developing The Solution

To help develop out solution we took all these into account and then created an initial idea to then get seen by an adult who could help see our solution from another perspective and gave the idea to just order a mirror online.